

UNIVERSITI POLY-TECH MALAYSIA

**XPLOREQUEST: A SECURE CROSS-PLATFORM
MOBILE EXPLORATION APPLICATION**

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APPLICATION**

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NOVEMBER 2026

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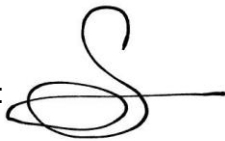
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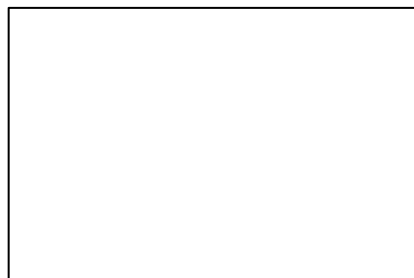
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Abstract

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Acknowledgements

1 INTRODUCTION

1.1 Introduction

Given the growth of mobile computing, outdoor adventure racing (a.k.a. Exploraces) has moved from a manual check-in method with paper to a digitally-based location method. This creates a high opportunity for digital cheating in a competitive event from tricking a GPS or simply taking a screenshot of a QR Code and using that to scan in. To address these areas of vulnerability, the purpose of this project is to introduce The XploreQuest Project and develop a secure, cross-platform mobile exploration application. XploreQuest is being developed using Expo (React Native) and Firebase and will utilize a combination of dynamic HMAC-SHA256 verification as well as offline caching to ensure the integrity and reliability of each event.

Chapter One presents the basis for the XploreQuest project. This chapter outlines the operational background of the client of an event coordinator, defines the core security issues and functional problems, identifies the overall objectives of the project, establishes the project and product scope of the project and product, identifies the target end users, and provides a summary of the structural layout of this report.

1.2 Project Background

In Malaysia, there are many different types of outdoor adventure races, geocaching games, and scavenger hunts that may be used for corporate or recreational team-building purposes. These activities can be considered part of a larger trend towards gamified location-based and physical activities, as noted by Hjorth & Richardson (2011). Typically, these types of events occur in large outdoor open spaces, such as Tasik Titiwangsa, public botanical gardens, and universities. Previously, event organizers relied heavily on manual checkpoint validation to manage these events; i.e., field marshals would physically provide stamps or signatures on paper cards that participants carry. However, when hundreds of people are spread out over large areas, there is significant potential for manual coordination (i.e., checking in and out) to become increasingly chaotic, slow, and error-prone, resulting in long wait times at busy checkpoints, bottlenecks during registration, and disputes regarding confirmation times. The need for planning and scheduling models that can dynamically route participants is evident from the negative impact on physical congestion at event locations due to the variability and unpredictability of these variables (Vansteenwegen et al. 2011).

This project aims to help solve the problem of inefficiency by creating a platform for the client (Dapo Awoknyee Resources) to use to facilitate the work that they do as a professional event management/coordinator firm located in Malaysia (KL Event Crew). KL Event Crew is managed by Puan Nurul Husna Binti Mustafa and provides services to large-scale events by arranging the hiring and

assigning of personnel to perform functions such as directing traffic, monitoring the start/finish line of a race, etc. This platform will allow the coordinator to map out the event boundaries, register teams, will have an automated check-in system, stagger team starting positions such that teams will not get backed up at check-in, and provide real-time leaderboards.

Most digital location-based and geocaching solutions fall into two categories: traditional (non-gaming) and gaming (casual). Traditional solutions do not have any significant security architecture, while casual solutions do. In addition, since XploreQuest is a Final Year Project for the Faculty of Computing & Multimedia at Universiti Poly-Tech Malaysia (UPTM), the producer has focused heavily on engineering for cybersecurity. To meet the requirements of competitive events, where there is a strong desire for cheating to gain competitive advantage, XploreQuest will also include defensive mechanisms to prevent the electronically assisted cheating from taking place. There are several other concerns regarding the application, all of which are related to its function as an electronic means to track the locations of participants in a competition; hence there are several requirements for both data and location privacy in accordance with the Personal Data Protection Act (PDPA) 2010 – Laws of Malaysia (2010), elevating XploreQuest as a legally compliant, research-supported, mobile system.

1.3 Problem Statement

Digital platforms for event check-ins at various physical locations are expanding rapidly, but current implementations of traditional check-ins contain serious vulnerabilities that hinder their security, equity, and dependability. The three categories of problems resulting from the assessment of location-based check-in systems can be related to the CIA Triad. Integrity vulnerabilities are associated with location and checkpoint apps that allow for cheating; confidentiality vulnerabilities exist for credential storage (e.g., in-person) and end user privacy; and availability and consistency issues exist due to the nature of outdoor environments (e.g., if an outdoor location loses power or has poor connectivity).

1.3.1 Lack of Security in the Checkpoint Validation Process

The standard check-in systems used in location-based gamification typically validate that a user has visited a location by using either a simple QR code or GPS coordinates. This means that as a competitor, you can easily circumvent physical locations and cheat at the checkpoint by taking a picture of the QR code at the location and sending it to a cheat team member thousands of miles away via WhatsApp or Telegram. The person who is thousands of miles away will then scan the image of the QR code and fraudulently check-in.

Likewise, those with access to developer mode on their mobile devices can use GPS spoofer applications to send fake location data directly to their mobile device's operating system, making it appear to the app that the user has visited the geofenced checkpoint without actually doing so (Seneviratne et al.,

2017). Numerous studies on mobile location services show that unvalidated GPS telemetry and static checkpoints are very susceptible to software-based location simulation tools therefore making it necessary to implement advanced telemetry validation and dynamic verification methods to ensure integrity (Seneviratne et al., 2017).

1.3.2 Insecure Caching of Session Credentials on Mobile Devices

Many mobile apps keep sensitive session, JSON Web Tokens and API keys in plain text and unsecured local storage (e.g., React Native `AsyncStorage`). On rooted or compromised devices, these credentials are probably easily extracted by malicious actors or side-channel applications resulting in session hijacking and undocumented access to APIs. Studies show that unencrypted device storage and plaintext key caches create unauthorized token extraction and identity leaks (Meng et al, 2018).

Additionally, many tracking applications continuously send and permanently keep their participants' historic GPS coordinates in a centralized database. This practice creates serious data privacy and regulatory compliance issues under the Personal Data Protection Act (PDPA) of Malaysia in 2010. Organizations that keep identifiable and permanent location histories of individuals when there isn't strict necessity are subject to legal penalties requiring a secure mobile architecture that provides ephemeral GPS telemetry, verifies location on-the-fly, and deletes coordinates immediately after the event.

1.3.3 Absence of Offline Support in Low-Connectivity Environments

Expansive parks, forest reserves, or rural areas with poor cellular (4G and 5G) service are the common venues for adventure racing. Most standard mobile apps are built with an online-first architecture that can only be used when they have a constant internet connection to send information to their backend servers. When an adventure racer is in a dead zone and cannot obtain a network connection, these applications will either freeze, crash or will not record the completion of a checkpoint. As a result of this loss of race progress, there will be corrupted leaderboard information and subsequently, a delay in race-day operations where race marshals will have to revert back to manually writing checks for the racers on paper.

Research conducted on mobile systems shows that a lack of robust offline caching and transaction ordering for synchronization results in significant instability and data inconsistencies in applications used in remote areas (Pothineni, 2024). Therefore, there is a need for offline resilient synchronization middleware that can store and queue transactions locally on an individual's device until the device is back online when they are able to upload their transactions automatically.

1.4 Project Objectives

To resolve the stated issues proposed in the previous section, the XploreQuest Project has been created with the following main objectives:

1.4.1 To develop a secure checkpoint validation system*

utilizing geofencing boundaries and dynamic HMAC-SHA256 verification mechanisms to prevent GPS coordinates spoofing and check-in forgery during outdoor adventure races.

1.4.2 To design a secure local storage architecture with token encryption

utilizing hardware-backed storage (such as iOS Keychain and Android KeyStore) via Expo SecureStore to protect session credentials and prevent token extraction on mobile devices.

1.4.3 To implement an offline synchronization queue for checkpoint scans

utilizing Mapbox local tile caching and a local SQLite database queue to secure and sync checkpoint telemetry once cellular network connectivity is restored.

1.5 Scope and Target User

Defining the scope and identifying the target user groups ensures that the project remains technically feasible and aligns directly with the operational needs of the client, KL Event Crew, without succumbing to feature bloat.

1.5.1 Project Scope

XploreQuest's main objective is to create and implement a mobile-based secure exploration application that can digitally secure and record the outdoor adventure events of KL Event Crew. The project will address major weaknesses in the current manual paper-based events and create a secure digital workflow. The client will have a cross-platform app (iOS/Android) built using React Native / Expo for the application. Firebase (Google's cloud suite) will handle user authentication, documents and real-time race synchronization across the backend of the application. Any type of physical infrastructure for example outdoor venue(s) that may require setup of a physical Wi-Fi router or local area network is not included in the scope of this project.

Cheating control, and protecting local IDs and keep the application available when network conditions are unpredictable have been the primary focus of this project's technical implementation. The project involves creating secure protocols for validating checkpoints using dynamic QR codes and geofences, and a velocity verification algorithm that uses GPS telemetry data to prevent coordinate spoofing. It also involves securing local session IDs using cryptographic storage on mobile devices with hardware-based security features and creating middleware for an offline-first database synchronization solution. The scope of the interface design includes working on interactive maps that help participants navigate the application, implementing role-based access control to restrict who can access data, and developing the backend security functions necessary to support these applications.

1.5.2 Product Scope

XploreQuest has developed a multi-tiered software application that includes two mobile applications (participant and crew) and an administrative website for use by event organizers. Participants will have

access to all of the features needed to participate in events such as registering for events and teams; viewing and using geofenced maps of the races, the locations of their current team's target checkpoints and their completed paths (but not disclosing actual coordinates of their checkpoints); scanning QR codes at the physical checkpoints; and tracking their team's progress towards completing the races in real time. The participant map will only show the team's current checkpoint and completed checkpoints and thus not reveal actual event coordinates to maintain the integrity of the competition.

In addition to the preceding functions, the crew/marshal functionality within the participant app will allow all physically deployed crew members/marshals to verify the arrival of teams at race locations as well as to validate course challenges and generate secure team-specific dynamic QR codes with cryptographically signed values (to prohibit screenshot disbursement). The third component of the XploreQuest application is an administrative web site used by the event organizer/administrator to configure the individual races; set geofence boundary values; manage team registrations; assign staggered/cyclical start times for teams in order to minimize the number of participants in close proximity to each other; and monitor live security alerts and real-time leaderboards as each of the events proceeds. The expectations for the performance of the overall application include achieving a System Usability Scale (SUS) score greater than 75 and a latency in achieving system synchronization upon restoration of the network of less than 5 seconds.

1.5.3 Target User

The XploreQuest system is designed for three distinct operational roles within the event ecosystem, each requiring a tailored interface to ensure safety, speed, and reliability:

1. **Event Participants (Team Leaders & Members):** This group consist of possibly three distinct types of people; college students, business workers, and members of the general population who participate in adventure racing. In order to conserve battery time and limit the traffic put on the cellular network, only one team member per team; the Team Leader will be the one that will download the app, operate the GPS geofencing tracker, and check-in at each checkpoint. All other team members will be able to directly observe the progress of the team on the Team Leader's screen.
2. **Field Crews (Checkpoint Marshals):** Set up physically at checkpoints are volunteers or part-time. The Checkpoint Marshals will utilize the Marshal View of the App to ensure that the teams have arrived at the checkpoint and completed their physical challenges and issue a dynamic QR code to verify their completion of the checkpoint's requirement. This dynamic QR code eliminates the need for Checkpoint Marshals to physically manipulate paperwork or use stamps or signatures.
3. **Event Administrators/Planners:** Use the web console to create the races, view live leaderboard standings, track where each of the Checkpoint Marshals are located, and receive alerts concerning possible cheating or geofencing violations that occur during the race.

1.6 Conclusion

Chapter 1 describes the foundational framework for the XploreQuest project. It describes how the nature of KL Event Crew, the client, created the operational environment and how the problems associated with security and functionality motivated the development of XploreQuest. By using the CIA Triad to outline the basic security, local data encryption and offline sync mechanisms, this chapter creates the parameters for establishing these functions and interfaces. The scope of the available objects and the participant/crew/administrative roles define where and how the system is to be used and what operational impact a user will experience in terms of addressing real-world vulnerabilities without creating unnecessary operational overhead.

Subsequent chapters in this report develop the process that will organise and capture all the individual phases of research, design and implementation. Chapter 2 presents a thorough literature review and is focused upon the conceptual structure of mobile security and geofencing as well as reviewing related gamified exploration platforms and identifying gaps in research that XploreQuest currently attempts to fill. Chapter 3 outlines the Agile Model and explains the requirement gathering techniques that will be used to elicit and analyse each of the system's attribute requirements. Chapter 4 presents an architectural design of XploreQuest's core database schema, implementation methodology and security framework, which includes the fundamental security framework and dynamic verification functions.

The report itself provides documentation related to the evaluation of the system and the final reflections upon it to validate its performance and security claims. In Chapter 5, the test plans and results from the functional, non-functional, and User Acceptance Test (UAT) that were completed during the live pilot, to validate location accuracy and check-in integrity, will all be discussed. Finally, Chapter 6 concludes this report by providing a summary of the overall achievements compared to the original objectives of the project, discussing project constraints of development, and providing recommendations for further enhancements. All the previous chapters can be referred to as two complete and comprehensive technical and academic resources for building and validating the XploreQuest platform.

2 LITERATURE REVIEW

2.1 Introduction

The literature review is an important part of as it demonstrates the theoretical foundation, provides an overview of current solutions and places the solution to be developed in relation to other academic literature or industrial applications. Chapter 2 will discuss the security, privacy, and architectural aspects of location-based mobile systems, focusing on the client's operational context at (Dapo Awoknyee Resources (KL Event Crew)).

The contents of this chapter are organized as follows. Section 2.2 offers an analysis of key technology areas that include: an analysis of cryptographically secure geofencing; the implementation of replay protection; the identification of target platforms and stakeholders; and the critical need for cryptographically secure location validation. Section 2.3 examines three current location-based services, Turf, Munzee, and Actionbound, by comparing those services' features, user interfaces, verification processes, and security postures in a detailed fashion. Section 2.4 then combines these analyses into a comprehensive matrix providing a comparison between the analyzed services and the proposed XploreQuest service. Finally, Section 2.5 will describe the weaknesses identified in the reviewed systems, and provide rationale for the security-focused architectural design decisions made in XploreQuest. To close out this chapter, section 2.6 summarizes the important points from the previous sections and discusses how they help to form the methodology that will be detailed in Chapter 3.

2.2 Investigation

For a secure, location-based mobile application, it's important to have a solid understanding of the technology (hardware and software) involved, environments in which your application will run, and protection from threats that affect location-based data. This will include understanding theory and practical implementation of security in geofences, replay protection mechanisms, people and entities involved in building the application (i.e., developers), and generic vulnerabilities associated with typical location-based services.

2.2.1 Understanding Cryptographically Secure Geofencing and Replay Protection

Virtual boundaries can be created around a geographic area using global positioning coordinates; this is referred to as geofencing. Typical geofencing services identify whether or not a user has entered the geofence by either measuring the distance from their device to the defined coordinate location, or by determining whether or not the coordinate location is contained within a polygon (polygon boundary containment algorithms). Physical barriers must be dynamically validated for high-stakes competitive events; this way only teams that are physically present within the geofence will be allowed access (Seneviratne et al., 2017). Regular GPS positioning systems can be used in ways that are not suitable for determining coordinates; examples include injection of coordinates, falsifying coordinates, and sending unauthorized coordinate information. Cryptographically secure geofencing solves these

problems by validating the information associated with a coordinate using math to ensure integrity, authenticity, and a timestamp.

Another important security priority for location-based check-ins would be implementing replay protection. A replay attack is when an attacker captures a legitimate check-in token or credential while it is being transmitted to the destination during a valid transaction and later uses the same token or credential to create an illegitimate check-in (OWASP Foundation, 2023). In outdoor adventure racing, for example, some participants may share images of valid QR codes for checkpoint validation with other teammates that are physically far away. By doing so, those illegally obtaining the QR code can use those images to create false checkpoint check-ins. Replay protection is achieved by ensuring that a check-in transaction can only be used at the time it was created; thus, they cannot be re-used. XploreQuest has created dynamic QR codes that are composed of the Hash-based Message Authentication Code (HMAC-SHA256) and can be digitally signed with a secret key by the verification system of the marshal and the server. The QR code will include the unique team identifier and a short-lived timestamp. If a participant uses a screenshot of the QR code after the expiration threshold has passed, the server-side validator will reject the transaction as invalid, therefore providing very robust replay protection.

2.2.2 Implementation Platforms, Stakeholders, and Target Environment

The targeted organization for the XploreQuest production is Dapo Awoknyee Resources (KL Event Crew) - Managed by Puan Nurul Husna Binti Mustafa. KL Event Crew is an event management company that recruits, trains, and deploys part-time crew members to coordinate large-scale outdoor exploration races (Exploraces) and corporate team building events, thus requiring a complete integrated mobile platform to replace the use of paper to manage events. In terms of implementing XploreQuest live, the platform is structured in three primary components; a Participant Mobile App, a Crew/Marshal Verification Website, and an Administrator Web Console.

The primary stakeholder groups for the system are as follows:

- 1 Event Administrators/Planners. Staff responsible for configuring events, setting up geographical checkpoints, creating geofence boundary coordinates, monitoring real-time security alerts, and managing the live leaderboard.
- 2 Event Participants. Teams of players who are navigating within the physical boundaries of the event via clues, using QR codes on each of the respective check-in points to confirm their arrival. To conserve battery life and reduce data traffic, the system has been designed using an architecture using single-tracker architecture where only the team leader is running the app.
- 3 Field Crews (Checkpoint Marshals). Staff located at physical checkpoints to verify team arrivals; evaluate results for the physical challenge areas; and create dynamic QR Codes to confirm their check-in.

2.2.3 Importance of Cryptographically Secure Location Verification

The use of standard location verification for mobile apps falls victim to a high amount of spoofing with bogus coordinates. There are many ways to simulate GPS coordinates via developer mode apps and Android mock location APIs, making them easy for developers to do. Also, participants can record completions without ever going to the actual checkpoint by copying, taking pictures of, or distributing static QR code systems. Using cryptographic location verification will guarantee that authentic transactions happen at each checkpoint by treating each completion as an individual, non-repudiable transaction using both GPS telemetry and dynamic signatures to verify all completions.

To prevent cheating, XploreQuest employs a dual-validation model. First, it implements a GPS telemetry velocity check algorithm: $v = \frac{\Delta d}{\Delta t}$

By calculating the straight-line distance (Δd) between consecutive check-ins and dividing it by the elapsed time (Δt), the system determines if the participant's movement speed exceeds a reasonable physical velocity threshold (e.g., maximum running or vehicle speed). If a player spoofs coordinates to cross a large distance instantly, the system flags the transaction as fraudulent.

The third aspect of the system's design is that it protects the end user's sensitive credentials resident on the device from unauthorized use. Mobile apps today typically have sensitive items, such as JWT session tokens and API keys, stored in memory structures in plain text without encryption. XploreQuest utilizes hardware-backed cryptographic storage (Expo SecureStore using the iOS Keychain and Android KeyStore) to encrypt sensitive items in their mobile applications by using keys that are stored exclusively in dedicated hardware.

The fourth aspect of the system's design is its ability to operate in an area with low-connectivity (e.g., forest reserves, parks, rural campuses). XploreQuest has created an offline-first synchronization queue that uses a local SQLite database to cache any checkpoint scans or telemetry data when the internet connection is not available. When the internet connection comes back up, the offline-first sync queue will ensure that all transactions created offline are queued securely and synchronized to Firebase Firestore in the order of their verified timestamps (Pothineni, 2024).

2.3 Related Works

To contextualize the development of XploreQuest, three existing location-based exploration, fitness, and geocaching applications are reviewed: Turf, Munzee, and Actionbound.

2.3.1 Turf

Turf is a mobile game that can be played with an outdoor physical experience and it's about controlling an area and being active at the same time. In this game, the map is divided into different groups with all the different simulated land uses or zones. You get a point(s) for the time you're in the zones you control and earn your points every 30 days under different reset periods.

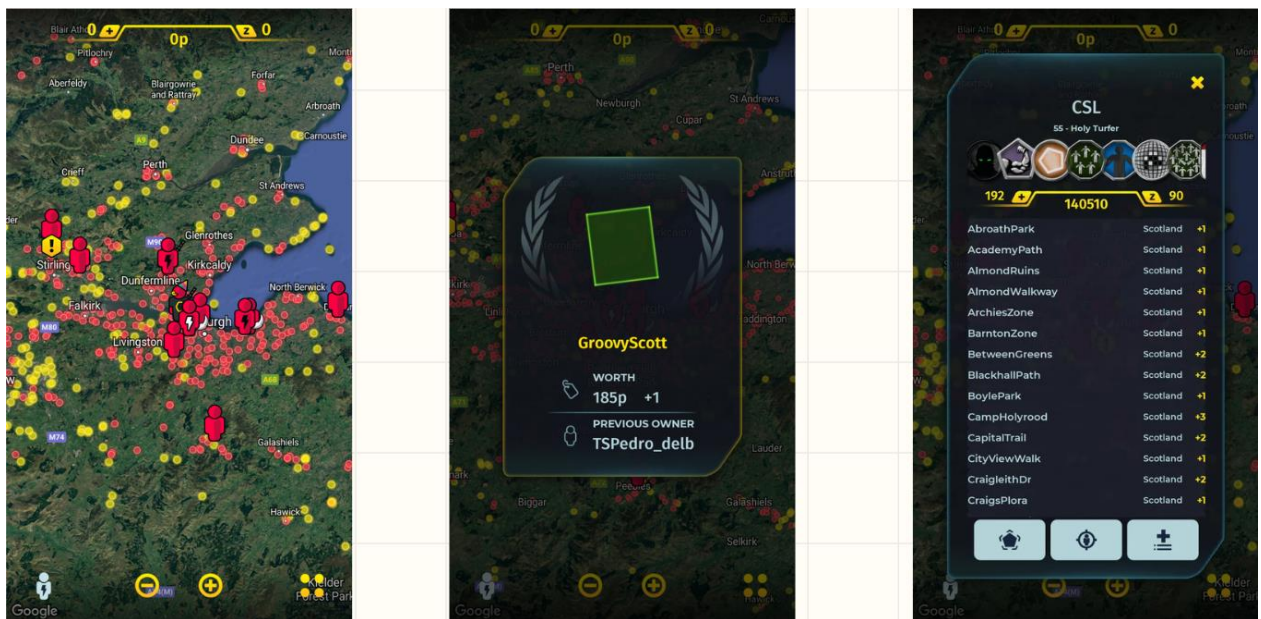


Figure 2.1 Turf user interface

Users & User Interface: The target users of Turf are people who are physically active; such as cyclists, runners, and walkers and people who want to play games just for fun. The program offers a live and interactive map; showing where you are, what zones are around you, and who the active users are in these zones.

Validation / Software / Hardware: Users of Turf validate their locations by using passive GPS (getting location by being near to that object's approximate geographic location). When users are in geofenced areas (like a certain "zone") for approximately 30 seconds after arriving in those areas they will receive points. Distribution of the Turf application is done using Native Applications (applications built specifically for Android and iPhone phones) and the hardware requirements is a cellphone device with a continuous Cell Network (CDMA, GSM, etc.) and GPS technology.

Since turf provides no security mechanism beyond server-side validation, it uses an application to compare the coordinates that a mobile device sends to the server-defined boundaries. As turf does not provide an encryption of its check-in verification or verification of device hardware, turf is highly

susceptible to GPS Spoofing, allowing users to spoof their location to claim territory without actually being present in the claimed area. Turf is a gamified exercise tracker but not secure enough for high-end, high-risk competition.

2.3.2 Munzee

The game of Munzee is played with real and virtual pins as a global treasure hunt/geocaching-type game that allows players to explore the world by finding items through both virtual and real-world means. Players can find a physical (sticker) QR code located in a public space, such as a park or urban area, which is called a Munzee or navigate to a coordinate where a virtual Munzee has been placed. Once found, the player who finds the item can receive points; likewise the player who placed (deployed) the item can receive points.

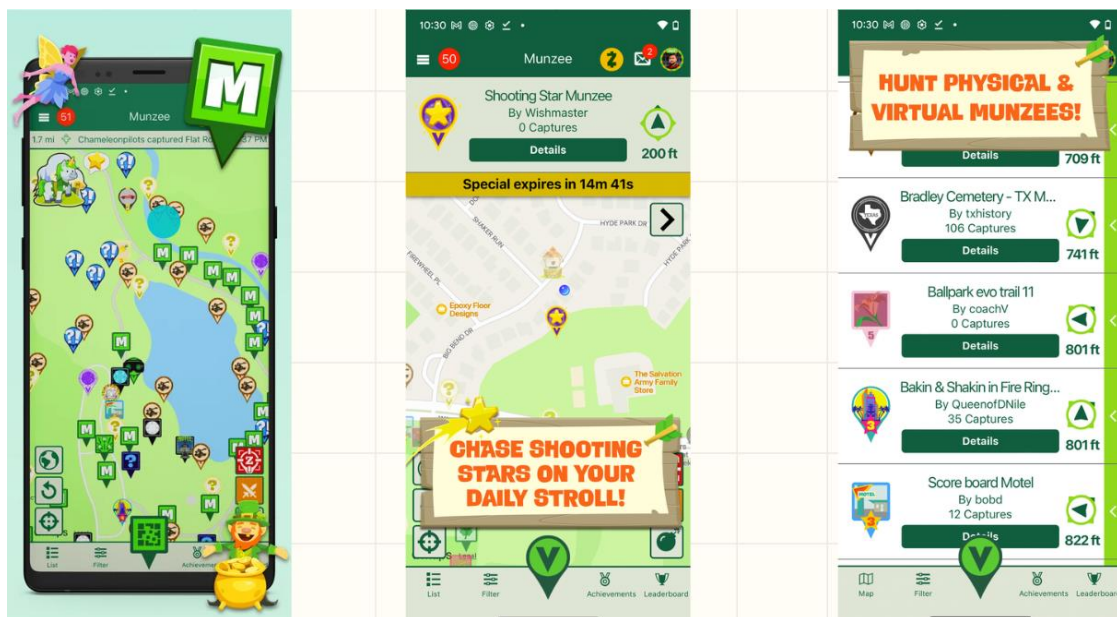


Figure 2 2 Munzee user interface

User and interface: The main target audience for Munzee are people interested in collecting type of geocaching, some more serious (collecting/mapping/data logging) than others, while others are simply casual explorers. Players can view the locations of real-world and virtual Munzees on an interactive global map showing the number of pins in each area of the world.

Verification, Software & Hardware: To verify the finding of a physical Munzee pin, the player must physically scan the QR code sticker using the Munzee application and also be within a certain radius (approx. 30) of the GPS coordinates of the pin. To verify the finding of a virtual pin, the player will automatically be verified when the GPS coordinate radius is within the Munzee geo-fence. Munzee is available for both Android and iOS smart phones; the Munzee application also requires that the player use a smart phone that has a camera, GPS and internet connection.

The security framework of Munzee is basic in nature. The QR codes that can be found on the stickers are printed as a statically fixed object, therefore these codes can be copied (photographed) or

distributed (shared) to players who are not present, allowing them to scan the munzee code from their home. There are a few basic GPS checks to ensure that a user is at the location of the QR code before scanning it; however, there are many ways for a developer to bypass these GPS checks by using mock-location apps. Thus, the use of Munzee successfully provides a method of collecting points by casual players; however, there are many ways for players to cheat on Munzee, which would not allow the use of Munzee in a professional event.

2.3.3 Actionbound

Actionbound is an interactive scavenger hunt creator that allows you to create digital scavenger hunts, mobile tours, and quizzes called "bounds". The organizer creates a bound using the web portal to establish the specific tasks to be completed and the specific location and/or quiz questions that must be completed.

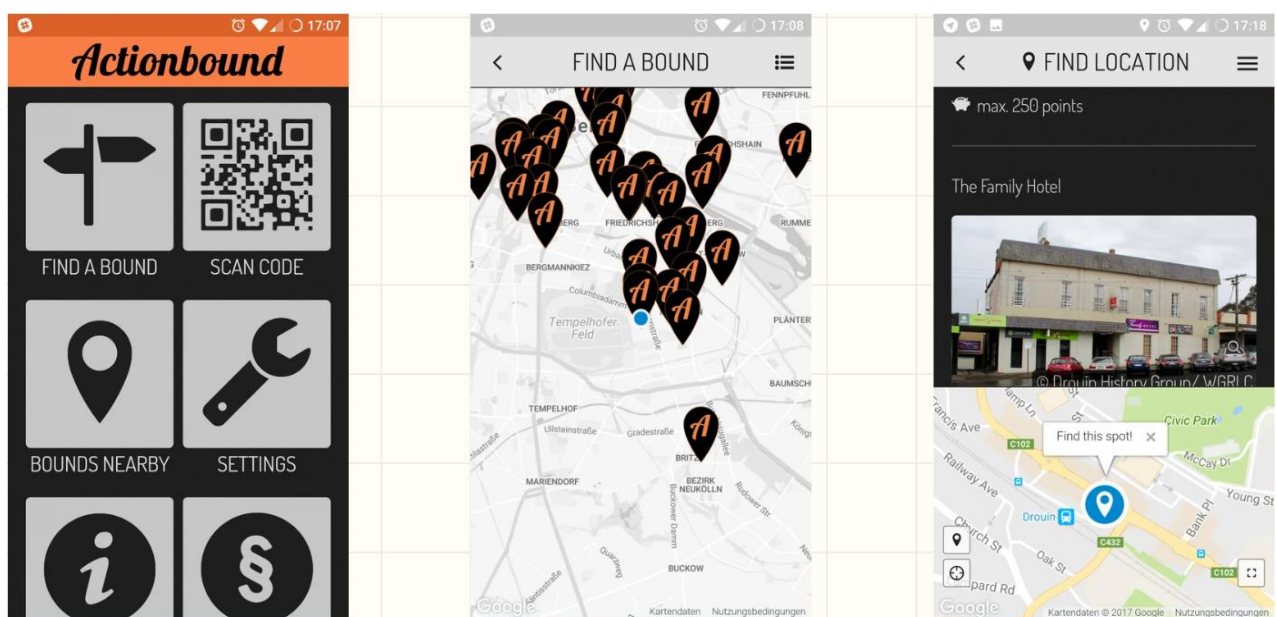


Figure 2 3 Actionbound user interface

User and Interface: Actionbound is intended to be used by educators, corporate trainers, museum coordinators, and tourists. The interface will present the users with a linear list of tasks, custom maps, etc., along with an interface for the compass to navigate to the next destination.

Verification, Software & Hardware: Verification of a Checkpoint is done in several ways. Players may verify they have completed a task at a Checkpoint by either scanning a static QR code, entering coordinates, answering a quiz, or providing evidence via media (photos/videos). The system utilizes offline boundaries where progress data and media are cached to the local device. The software solutions available are mobile applications for Android/iOS, plus a web-based management console for administrative functions. All players require a mobile device (with GPS), storage memory, camera, and microphone functionality, to validate tasks.

Security and Outcomes: Actionbound has a few basic verification rules for Checkpoint use. Verification includes: 1) checking for QR code string match against defined boundaries, and 2) checking geolocation of a device is in proximity of the boundary. Problems exist with both QR code validation and GPS for Checkpoints since QR codes are static, and GPS cannot assure correct validation. This opens the door for participants to provide screenshots of QR codes as false verification that they were at a Checkpoint. While Actionbound is an excellent tool for corporate team-building and education, it does have limitations as it lacks dynamic cryptography and velocity telemetry validation checks, and there is no secure keystore for credentials, thus leaving the door open to cheating based upon participants using actual media to determine who participated competitively.

2.4 Comparison

To systematically compare Turf, Munzee, Actionbound, and the proposed XploreQuest solution, the operational and technical characteristics of each platform are aggregated in Table 2.1.

Table 2 1 Comparison of Existing Project

Criteria / Project	Turf	Munzee	Actionbond	XploreQuest
Features	Territory capture, monthly rounds, global leaderboards, virtual zones.	Scavenger hunting, virtual/physical pins, in-game item store, clan battles.	Custom quest creator, offline bounds, quizzes, multimedia missions.	Secure dynamic QR scanning, offline queue syncing, real-time marshal dashboard.
User	Individual fitness gamers.	Individual collectors and casual explorers.	Educators, corporate groups, and casual tourists.	Event Participants, Event Marshals, and Event Organizers
Navigation / Interface	Real-time map with active player/zone markers.	Map with pins (physical vs. virtual markers).	Custom maps, list of quests, and directional compass.	Interactive map (Leaflet/Google Maps) with responsive web portal for marshals.
Pickup Verification	Passive GPS proximity (staying in a zone for ~30 seconds).	Scanning static QR code stickers OR passive GPS range checks.	Scanning static QR codes, GPS coordinates, or uploading media proofs.	Dynamic QR code scanning (time-sensitive signature) + Server-side GPS verification.
Software	Android & iOS mobile apps.	Android & iOS mobile apps	Android & iOS mobile apps	Android & iOS mobile apps
Hardware	Smartphone (GPS, internet connection)	Smartphone (GPS, camera, internet connection).	Smartphone (GPS, camera, local storage, microphone).	Smartphone (GPS, camera, SecureStore, local SQLite database for offline queue).
Security Elements	Basic server validation (susceptible to GPS spoofing).	Basic location validation (highly vulnerable to QR sharing/screenshots).	Basic GPS/QR match (vulnerable to QR screenshot sharing).	Multi-layer security: Dynamic HMAC-SHA256 signatures, GPS velocity telemetry checks, Expo SecureStore.
Result	Fitness tracking and competitive zone control.	Casual point collection and global player ranking.	Educational tours, team building, and custom gamified events.	Secured checkpoint validation preventing spoofing, screenshot sharing, and event cheating.

2.5 Discussion

As illustrated in Table 2.1, the three comparative studies show significant gaps in the security and operations of the current location-based apps. These apps (Turf, Munzee, and Actionbound) use mobile GPS and QR code verification for both physical exploration (exploration) and for creating outdoor games (gamification) but have designs based on low-risk environments where cheating would not have any effect on the material rewards or corporation rankings of the participants.

In particular, all three reviewed platforms are based on either static QR code verification or passive GPS verification. For Turf, participants may take advantage of the zone-capture feature by sending fake coordinates using common bogus location APIs. The design of both Munzee and Actionbound does not take into account the reliance on either printed or still images of QR codes, both of which represent vulnerabilities. Once a QR code photograph exists, it can be used by users from all over the world; so users can fake their presence from remote locations. Further, none of these apps have a method that stores credential data securely on a device, making it easy to obtain an access session key by illegally accessing an API of a rooted or jail-broken device.

Moreover, Turf and Munzee's dependence on a constant and active network connection can contribute significantly to unreliability. There are many network disconnections within KL Event Crew's managed outdoor environments. While it is possible for Actionbound users to cache their bounds offline, there is no way for a user to ensure that they are unable to tamper with their device's clock in order to falsify their race times, nor is there a way to verify that the user's completed checkpoints will remain intact prior to an online connection being established.

XploreQuest directly addresses these vulnerabilities. By integrating dynamic HMAC-SHA256 signatures generated in real-time by marshals, the system prevents screenshot sharing. The verification code becomes invalid after a brief operational window, rendering screenshots useless. To prevent GPS spoofing, the system combines geofencing checks with a GPS velocity telemetry check algorithm ($v = \Delta d / \Delta t$). By calculating the velocity of travel between consecutive checkpoints, the system flags and rejects instantaneous coordinate jumps.

Expo SecureStore enforces the security of access credentials at rest on the device (using iOS Keychain and Android KeyStore), preventing any key-extraction. Finally, the offline synchronization middleware uses a local SQLite database to log check-in transactions with securely and cryptographically hashed timestamps. Once cellular access is restored, transactions are synchronized in order, ensuring leaderboard consistency and operational reliability for KL Event Crew.

2.6 Conclusion

This comprehensive literature review in Chapter 2 provides both an academic and technical foundation for the application of XploreQuest. Chapter 2 also includes an investigation into secure geofencing boundaries, protection against replay attacks using HMAC-SHA256, and the validation of telemetry data; and as a result, the chapter has not only determined what security controls will be necessary to protect competitive adventure events but also identified three major flaws in existing location-based platforms, including Turf, Munzee, and Actionbound: susceptibility to coordinate spoofing, the ability to duplicate static QR codes, and lack of local secure storage of data.

Furthermore, the comparisons and discussions made in this chapter highlighted that while casual gaming systems have been able to successfully meet their recreational purposes, they lack the reliability and security that KL Event Crew requires. To mitigate the deficiencies found, XploreQuest includes a multi-layered security model comprised of dynamic HMAC verification, velocity checking of telemetry, secure hardware credential storage, and an offline-first design for synchronizing using SQLite.

This chapter sets out the system's design parameters by detailing security requirements and verifying the project's architectural decisions. Chapter 3 - Methodology, describes how agile methodology, data collection techniques and requirement specifications are used to design, develop and implement XploreQuest.

3 METHODOLOGY

3.1 Introduction

In the methodology chapter, the reader learns about the systematic approach to the investigating, analyzing, designing, developing, and validating of the application XploreQuest. A software development life cycle (SDLC) is selected to meet the basis of the design and development objectives based on project constraints, as well as yielding a robust, secure, and user-friendly product (Beck et al, 2001). The Agile development methodology, as well as the Scrum development framework are discussed in regards to specification types (i.e., qualitative & quantitative data-gathering methodologies), hardware & software specifications, analyzing data, and creating conceptual models for guiding use of an iterative methodology for development.

The chapter is arranged as follows: Section 3.2: Justifies the selection of the Agile methodology and describes how the Scrum framework was used; Section 3.3: Describes the iterative phases of the Agile-Scrum model, from planning to final implementation and review; Section 3.4: Requires techniques for gathering requirements (i.e., using questionnaires and interviews) to specify functional, non-functional, and system requirements derived; Section 3.5: Contains analyses of gathered data, including survey results and theme analyses of recorded interviews, together with a use case model and system flow charts illustrating the interaction between actors; Section 3.6: Concludes the chapter providing a summary of these methodologies' translation to design parameters for Chapter 4.

3.2 Agile and Scrum Methodology

The process of developing the XploreQuest app is done using an iterative and incremental software life cycle method and the Agile Methodology; specifically through a process called the Scrum Framework. The Agile Methodology provides an iterative and incremental software development approach that focuses on continuous collaboration between teams, the ability to change in real time, continuous testing of the product, and the ability to quickly respond to changing product requirement (Schwaber & Beedle, 2002). The Scrum Framework, which is a subset of Agile, organizes development in fixed duration cycles called sprints (2-week intervals of time) (Schwaber, 1997) in order to provide an environment that allows the developer to continuously receive feedback from stakeholders and to continuously develop and enhance features in the application. This creates a great tool for developing mobile applications such as XploreQuest, because there are many variables that play a part in determining the usability of the application for the end users (the User Interface, security mechanics, backend services, etc.) and all of these variables must be tested, and modified according to various factors.



Figure 3 1 Agile Methodology Diagram

The methodology used to develop this application was based on three primary principles:

- 1 Flexibility and Adaptability: The requirements of location-based security and anti-cheat validation for this project are very complex and have required considerable modification throughout the design of the application due to consultation with the client, Dapo Awoknyee Resources (KL Event Crew).
- 2 Iterative Development: The majority of the core features of the application (geofenced maps, offline synchronization, and dynamic QR Verification) were developed, tested, and modified through very rapid development cycles.
- 3 Continuous Testing and Quality Assurance: The system has been subjected to security validation checks (mock-location testing), and the verification of database transactions have been conducted during every sprint in order to check for exploits within the system.

3.2.1 Scrum Roles

In order to incorporate Scrum into my Final Year Project, the standard roles associated with Scrum have been assigned like this:

Product Owner: Puan Nurul Husna Binti Mustafa (the client representative for Dapo Awoknyee Resources; aka KL Event Crew). As Product Owner, Puan Nurul defined the business requirements, prioritized features in the Product Backlog, and evaluated increments (development work) for meeting the operational needs of the event crew.

Scrum Master and Developer: The student developer. As Developer, they wrote the code, configured databases, and performed unit testing. As Scrum Master, they ensured that everyone adhered to the Scrum process and were managing the task boards, including removing technical blockers (e.g., emulator coordinate overrides).

Three primary Scrum artifacts were used to manage the development process:

- 1 Product Backlog: A comprehensive prioritized collection of features, user stories, and security validations that form the basis for the Product Backlog. The Product Backlog is specifically referenced in the requirement section at 3.4, and revisions have been made based on client feedback throughout the process.
- 2 Sprint Backlog: A set of user stories selected from the Product Backlog for completion during a specific sprint. For example, the acceptance criteria for geofencing validation were established as separate user stories and delivered over separate sprints.
- 3 Increment: The completed, working version of the XploreQuest application delivered at the conclusion of each sprint (e.g., an Android Package File (APK) or Expo client).



Figure 3 2 Scrum Diagram (Stacey Ackerman 2022)

3.2.3 Scrum Ceremonies

To ensure the project remains structured and realized throughout the entire project lifecycle, a variety of ceremonies (or Agile-Scrum processes) were adjusted as follows:

- 1 Sprint Planning: At the start of each sprint, this ceremony is used to set clear goals for that sprint, prioritize and review the order/priority of the backlogged items and to estimate the amount of effort needed to implement those items.
- 2 Daily Scrum: This was adapted to be a personal daily record/log of the developer's progress and tasks completed from the previous day, planned focus for the current day's work and any tasks blocking progress.
- 3 Sprint Review: This meeting is held with the client representative (the Product Owner) to review the completed functional software increments (such as the dynamic QR generator and geographical boundaries of the geofence) and to receive validation that everything has been met.

- 4 Sprint Retrospective: The retrospective meeting is held at the conclusion of each sprint to review and evaluate the technical workflows used during development, to evaluate bugs found (such as memory leaks found while using background location tracking functions) and to determine methods for improving development processes for the next sprint.

The iterative process followed in the Agile-Scrum development model can be seen in Figure 3.1 and illustrates how requirements and designs flow to code, test, and review the implementation.

3.3 Phases in Agile-Scrum Methodology

The development lifecycle of XploreQuest is executed across five distinct phases, as detailed below.

3.3.1 Phase 1: Planning

The planning phase included defining the scope of the project, research objectives, and the boundaries of operation of the application. The product backlog was created by consulting with customers about the order in which tasks should be performed. In this planning phase, two significant security issues were highlighted: 1) the sharing of static QR codes; and 2) GPS spoofing; and target deadlines were established along with the set of milestones.

3.3.2 Phase 2: Design

During the design phase, the backlog was transformed into technical specifications. The specifications were based on a defined security framework, wireframe designs created in Figma, and the database structures were modeled for both Firebase Firestore and local SQLite. In addition, use cases were organized, and flow charts were created to illustrate the interaction of the crew, the system and the key participants.

3.3.3 Phase 3: Development

In phase 3, application logic was developed to create a cross-platform mobile client built with Expo (React Native), Leaflet/Google Maps for geofenced mapping, Expo Camera for scanning, and Expo SecureStore for encrypted local storage. Backend Architecture was developed with Firebase for authentication, real-time database tracking using Firestore, and media storage using Firebase Storage.

3.3.4 Phase 4: Testing

During the entire development process, continuous testing was performed. Unit tests were performed using Jest to test geofenced coordinate calculations and HMAC payload signatures. Security testing was performed to check for detection and blocking of mock-location apps and spoofed coordinates. Offline testing was performed to simulate network drops to confirm that the SQLite cached transaction queue would sync once the mobile device regained cellular service.

3.3.5 Phase 5: Deployment and Review

The final phase consisted of deploying serverless cloud APIs and Firestore databases into the production environment. A staging version of the mobile app was given to the client via Expo Go, and a review was conducted to gather client feedback, measure system usability, check log latency, and patch minor bugs prior to the final signing off on the project.

3.4 Requirement

This section describes the various techniques used to gather information for developing the XploreQuest app and the functional, non-functional, and system requirements.

3.4.1 Data Gathering Techniques

To gather complete data to create strong requirements, the developer used a mixed-methods research design that involved both quantitative and qualitative approaches.

3.4.1.1 Questionnaire

A quantitative online questionnaire was created and distributed using Google Forms to youth, university students, and members of the public who have participated in outdoor adventure races, scavenger hunts, or exploration-based events. A minimum of 50 people were surveyed in order to obtain a representative sample. The survey consisted of four categories: demographics, reliability of the network in an outdoor environment; awareness of location-based cheating methods (like GPS spoiling or QR screenshots); and the perceived usefulness of proposed features in the XploreQuest app.

3.4.1.2 Interview

A qualitative, structured interview was conducted via online communication channels with Puan Nurul Husna Binti Mustafa, the administrator of Dapo Awoknyee Resources (KL Event Crew) on 5 June 2026. The interview's objective was to determine the operational workflows and identify pain points regarding physically registering at checkpoints, and refine the app's flow-control and anti-cheating systems.

3.4.2 Functional Requirement

Functional requirements define the core capabilities and services that the system must provide. Based on stakeholder consultations, the functional requirements are summarized in Table 3.1.

Table 3 1 XploreQuest Functional Requirements

Req ID	Feature Category	Description	Priority
FR-01	Dynamic QR Generation	The Crew App must generate a cryptographically signed, time-sensitive (30s) QR code locked to a selected team to prevent	High

		screenshot sharing.	
FR-02	Checkpoint Skip Logic	Participants must be allowed to bypass busy checkpoints, marking the checkpoint status as "Pending" and allowing them to complete other tasks first.	High
FR-03	Finish Line Gatekeeper	The system must block final check-in scans if any checkpoints are flagged as "Pending" and display warning alerts to the team leader.	High
FR-04	GPS Geofence Validation	The system must automatically check if the participant's location falls within the geofenced polygon boundary before allowing a scan.	High
FR-05	GPS Velocity check	The backend must calculate travel velocity ($v = \Delta d / \Delta t$) between consecutive scans to detect and block mock-location spoofing.	High
FR-06	Offline Sync Queue	The mobile client must cache checkpoint scans in a local SQLite database during network dropouts and auto-sync when connection returns.	High
FR-07	Countdown Timers	The app must implement countdown timers for checkpoints and automatically calculate time penalties (e.g., +5 minutes) for expired tasks.	Medium
FR-08	Photo Verification	The mobile client must support uploading live photos at checkpoints as additional evidence of task completion.	Medium
FR-09	Live Leaderboard	The administrative web console and participant apps must display real-time rankings, scores, and safety alerts.	Medium
FR-10	Session Recovery	The application must recover the exact active state of the race if the user logs back in after a battery failure or app crash.	Medium

3.4.3 Non-Functional Requirement

Non-functional requirements specify the criteria used to judge the operation of a system, focusing on quality attributes, safety, and constraints.

- 1 Security: Every API request will need to be validated with a JSON Web Token (JWT). All sensitive local credentials (tokens, secret keys) must be stored in a hardware-backed storage location (Expo SecureStore that accesses iOS Keychain/Android KeyStore). The validation secret key must dynamically rotate to mitigate the risk of replay attacks.
- 2 Performance: The maximum allowable server response time for validating a checkpoint should be less than or equal to 1.5 seconds for a standard cellular connection. Background GPS tracking must be optimized for battery performance with a maximum of 8% battery drain during the continuous tracking of a race.
- 3 Usability: The user interface should be optimized for mobile use with legibility considered that includes using large fonts; providing clear progress indicators; and having visual representations of

completed activities (✅) for visibility when outside. The color scheme should use high-contrast colors to be viewed when outside in direct sunlight.

- 4 Reliability (Offline Resilience): The application must work without any loss of data in a no cell coverage area by caching up to one hundred (100) entries in the queue locally. Auto-synchronization must automatically synchronize the queue entries that were cached at the time of the first detection of a cellular network without having any manual synchronization done by the user.

3.4.4 System Requirement

To develop, run, and test the XploreQuest application, specific hardware and software tools are required.

3.4.4.1 Hardware Requirement

The hardware requirements are categorized into the developer's workstation specifications and the target users' mobile device standards.



Figure 3 3 Honor Magicbook X15

Table 3 2 Developer Hardware Specifications

Specification	Description
Workstation Model	HONOR MagicBook X 15 (BBR-WAX9)
Processor	Intel® Core™ i3-10110U CPU @ 2.10GHz
RAM	8.00 GB
Storage	256 GB SSD
Graphics Card	Intel® UHD Graphics
System Architecture	64-bit Operating System, x64-based Processor
Operating System	Windows 10 Home Single Language

3.4.4.2 Software Requirement

The software tools utilized during the development, testing, and deployment of XploreQuest are detailed in Table 3.3.

Table 3 3 Software Development Requirements

Software Category	Tool / Platform	Purpose in XploreQuest
Operating system	Windows 10 Home	Host environment for coding, emulation, and documentation.
Code editor (IDE)	Visual Studio Code (VS Code)	Development environment for writing code and debugging.
Mobile framework	Expo (React Native) SDK 49+	Development of cross-platform native iOS & Android applications.
Programming language	JavaScript / TypeScript	Implementation of application logic and backend validations.
Backend & hosting	Google Firebase	Firebase Auth, Firestore real-time database, and cloud functions.
Database management	Firebase Console / SQLite Viewer	Managing online collections, security rules, and local database tables.
Version control	Git & GitHub	Code repository backup, version tracking, and release management.
Emulation / testing	Android Studio Emulator / Expo Go	Emulating Android client and running physical on-device testing.
UI/UX design tool	Figma	Drafting UI wireframes, flows, and graphics.
Office productivity	Microsoft Word / Google Forms	Drafting documentation, slides, and distributing surveys.

3.5 Analysis

This section analyzes the qualitative and quantitative data gathered from the stakeholders, followed by the conceptual system use cases and flowchart designs.

3.5.1 Data Gathering Analysis

The analysis of questionnaires and interview responses provided empirical support for the application's design and confirmed the critical need for secure, offline-capable location services.

3.5.1.1 Questionnaire Analysis

The quantitative survey gathered responses from a total of 57 valid respondents after removing incomplete entries. The questionnaire was structured into four sections to analyze demographics, connectivity challenges, cheating awareness, and the perceived utility of the proposed XploreQuest features.

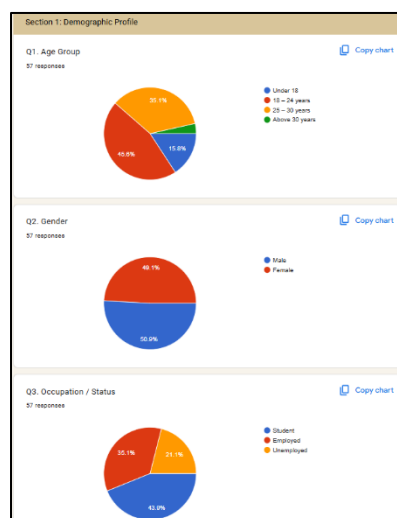


Figure 3.4 Section 1: Demographic Profile

- The demographic data indicates that the primary target audience consists of young, tech-savvy individuals who are highly likely to engage with mobile exploration applications. Table 3.4 summarizes the demographic characteristics of the survey sample.

Table 3.4 Respondent Demographic Profile

Demographic category	Profile group	Frequency (n)	Percentage (%)
Age group	Under 18 years	9	15.79%
	18 – 24 years	26	45.61%
	25 – 30 years	20	35.09%
	Above 30 years	2	3.51%
Gender	Male	29	50.88%
	Female	28	49.12%
Occupation / status	Student	25	43.86%
	Employed	20	35.09%
	Unemployed	12	21.05%

From the data presented in Table 3.4, most (80.70%) of the participants were between the ages of 18 and 30 years (with 45.61% at ages 18–24 years and 35.09% at ages 25–30 years). In terms of overall respondents, there were more respondents enrolled as students (43.86%) than those who were employed (35.09%). These profiles illustrate the standard demographic characteristics for both university students and young adults associated with youth-focused organisations that are normally involved in outdoor activities organised by KL Event Crew.

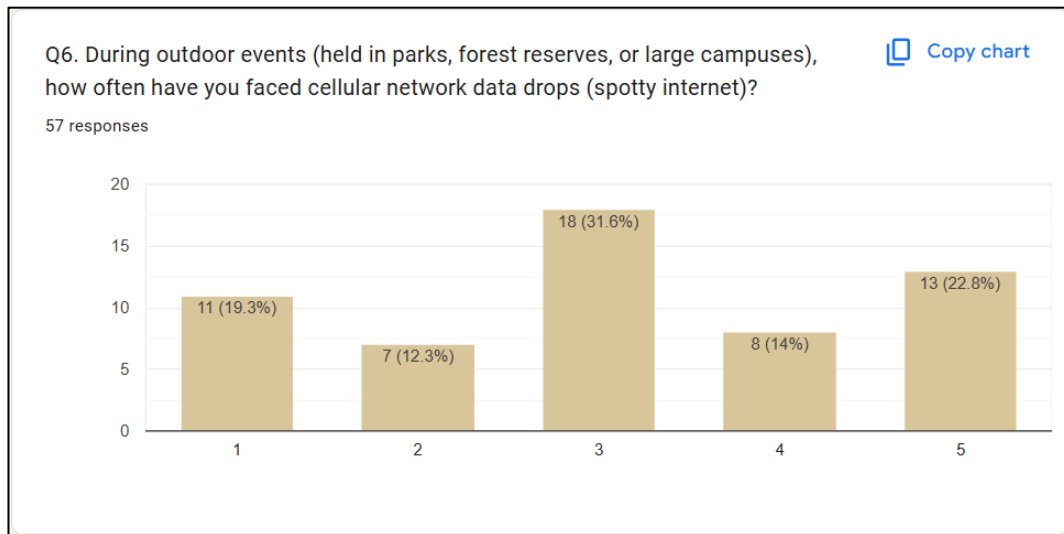


Figure 3 5 Problem with cellular data drops

- Outdoor events are often held in large areas such as university campuses, parks or forests where cellular coverage varies widely based on location. Participants were asked to rate how often they had problems with cellular data drops on a scale of 1 (never) to 5 (very often). Results of the survey show that: 36.84% of participants (21) reported having either frequently or very frequently had issues with having dropped cellular signals (ratings of 4-5). 68.42% of participants (39) had either sometimes or very often (ratings of 3-4-5) experienced dropped cellular signals. 19.30% (11) of participants said that they had never experienced dropped cellular signals, which indicates the inconsistency of cellular service in outdoor settings.

Since a mobile checkpoint application needs constant access to the internet in order to track arrivals in real time, the statistical evidence indicates that more than 67% of participants will experience disruptions. This provides explicit evidence to support the inclusion of FR-06: Offline Sync Queue, allowing mobile clients to store completed checkpoints locally within an SQLite database and automatically sync once a cellular connection is restored.

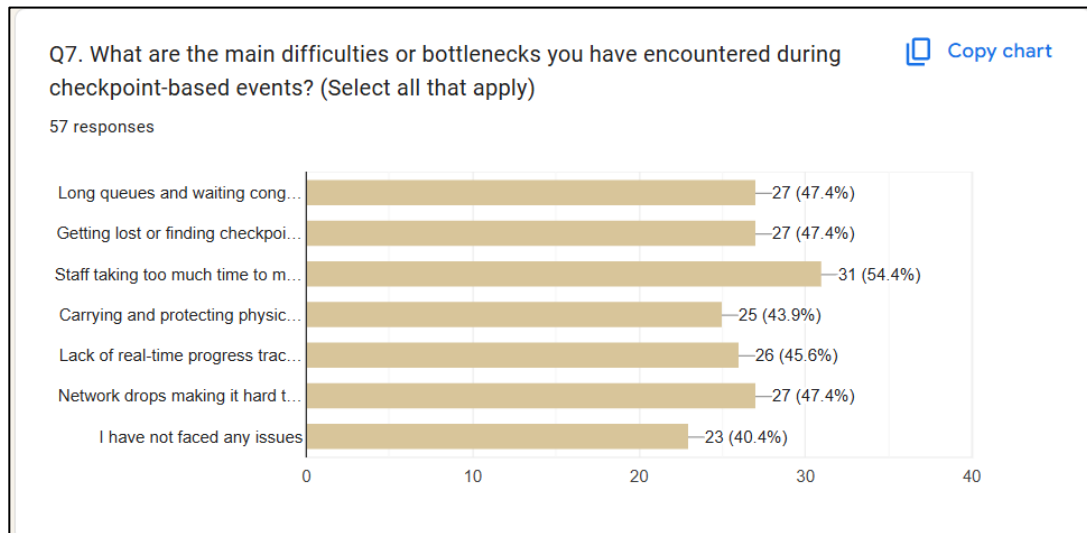


Figure 3 6 Main difficulties

- 3 Respondents encountered several difficulties and bottlenecks related to physical checkpoints or events based on a section 2 (Q7) on 10-12-07. Respondents could select multiple answers for this question from which we compiled answers for Section 2 (Q7), into Table 3.5 in descending order from frequency.

Table 3 5 Encountered Checkpoint Difficulties and Bottlenecks

Difficulty / bottleneck category	Frequency	Percentage
Staff taking too much time to manually record or sign off team arrivals	31	54.39%
Long queues and waiting congestion at physical checkpoints	27	47.37%
Getting lost or finding checkpoints difficult to navigate to	27	47.37%
Network drops making it hard to access event information	27	47.37%
Lack of real-time progress tracking or live leaderboards	26	45.61%
Carrying and protecting physical, paper-based stamp cards/clues	25	43.86%
I have not faced any issues	23	40.35%

Table 3.6 reveals several major usability issues: Sign-off by hand creates bottlenecks and staff manual entry requirements are creating congestion in the process (54.39% indicated that manual recording is a major bottleneck due to congestion) supports the need for FR-02: Checkpoint Skip Logic, which will support teams in skipping over congested checkpoints when completing others and returning later. Tracking was stated by 45.61% of respondents as an issue (lack of progress tracking and live leaderboard) supports need for implementing FR-09: Live Leaderboard (to visually show current standings of teams and status of their events). Carrying paper stamp cards is inconvenient for 43.86% of respondents supports need for a fully digitized mobile-driven validation portal.

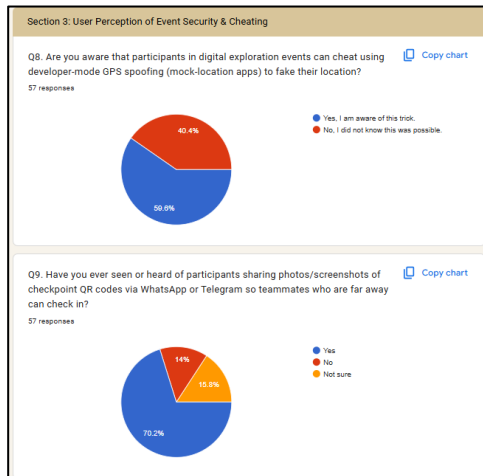


Figure 3 8 Section 3: User Perception of Event Security & Cheating (Q8-Q9)

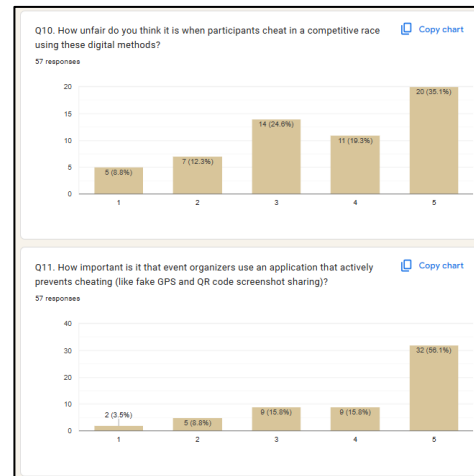


Figure 3 7 Section 3: User Perception of Event Security & Cheating (Q10-Q11)

4 Location Cheating Awareness and Security Perceptions

Section 3 of the survey examined the participating individuals' knowledge of cheating techniques associated with geographic information systems: GPS Spoofing Awareness (Q8): 59.65% of the individuals surveyed (34) are knowledgeable of individuals who utilize developer mode GPS spoofing (i.e. to fake the individuals' location), in order to cheat at events conducted digitally. QR Code Screenshot Sharing (Q9): 70.18 % of the individuals surveyed (40) have personally seen participants of events that take part in staging check-ins remotely through screenshots sent via WhatsApp or Telegram. Perceived Unfairness (Q10): The majority of the individuals surveyed perceived "digital cheating" via a rating system of 1 (Not Unfair) - 5 (Extremely Unfair) with 54.39% of the individuals surveyed (31) indicating they perceive digital cheating as being either highly or extremely unfair (rating 4 or higher). Importance of Active Security Prevention (Q11): The majority of the individuals surveyed indicated that the use of an application by an event organizer that prevents cheating or cheating from occurring is highly (rating 4 or higher) important to them in order to prevent cheating (i.e. 71.93 of the individuals surveyed, 41).

GPS spoofing and sharing screenshots of QR codes is a commonly-known method to undermine the integrity of fair competition. They have confirmed that both static QR codes and non-validated GPS checks are inadequate. As such, the following features should also be implemented:

- i. FR-01: Dynamic QR Generation: Create QR codes that will only last for 30 seconds, generate a new one every 30 seconds and create QR codes with a cryptographically signed HMAC payload, so if someone takes a screenshot of a QR code it will not work later.
- ii. FR-04: GPS Geofence Validation: A participant will not be able to check in unless their physical location is within the defined geographical area of the event.

- iii. FR-05: GPS Velocity Check: Check how fast a participant is travelling based on the distance between checkpoints to flag unusual travel speeds where coordinates may have been faked or altered.

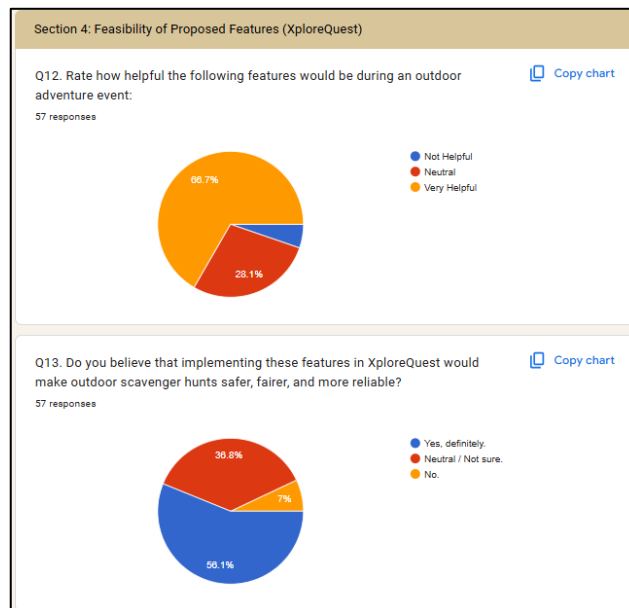


Figure 3.9 Section 4: Feasibility of Proposed Features (XploreQuest)

5 Proposed Feature Utility and System Trust

The evaluation of the utility of the features was conducted in Section 4: Helpfulness of Features (Question 12): 66.67% of participants (38) rated skip logic, offline sync., progress checklists, & dynamic QR verification as "Very Helpful"; 28.07% were Neutral (16); 5.26% felt the features were "Not Helpful " (3).Credibility of the System (Question 13): 56.14% of participants (32) believed implementing features in XploreQuest will create safer, fairer, and more reliable outdoor scavenger hunts; 36.84% were Neutral/Unsure (21).Qualitative Input (Question 14): Suggestions provided in the open-ended section for improving the security of the system included: device-binding controls (locking participant login sessions to a single, unique hardware ID to preclude participant account sharing) and mock-location detection APIs. These were added as Non-Functional Requirements in Section 3.4.3.

3.5.1.2 Interview Analysis

The thematic analysis of the WhatsApp interview with Puan Nurul Husna Binti Mustafa (KL Event Crew) revealed four key themes that would inform the development of this system:

- Control of Race Flow & Reduction of Bottlenecks: The client believes that manual queues at checkpoints detrimentally affect the experience of participants, therefore in order to alleviate this, the system should enable participants to bypass busy stations and first complete easier

checkpoints whilst disallowing final check-ins should there be any unfinished tasks.

- b) Time Limits, Alerts & Penalties: The client proposed the implementation of countdown clocks. A time penalty (adding, for example 5 minutes to the total score) should be applied to teams for exceeding the time allocated for a task, and exceeding the time limit for all tasks will classify the team as DNF (Did Not Finish) in the program.
- c) UI Feedback and Progress Tracking: Clear notifications must be provided to advise players of any outstanding tasks, therefore the developer would recommend the use of checkmarks (✅) and progress bars for each player as they relate to completion rates for tasks in the dashboard.
- d) Dynamic QR-scanning to Prevent Cheating: The client advised that the most common form of cheating for this event would be using an image of a QR code. The system would address this through the use of dynamic QR codes generated on the marshal's screen that would expire after 30 seconds of generation and would also require a GPS radius check for them to be completed as valid by the participant.

3.5.2 Use Case Model

The system design defines the actors and use cases necessary to implement these rules. There are four distinct actors (three human roles and one automated system actor):

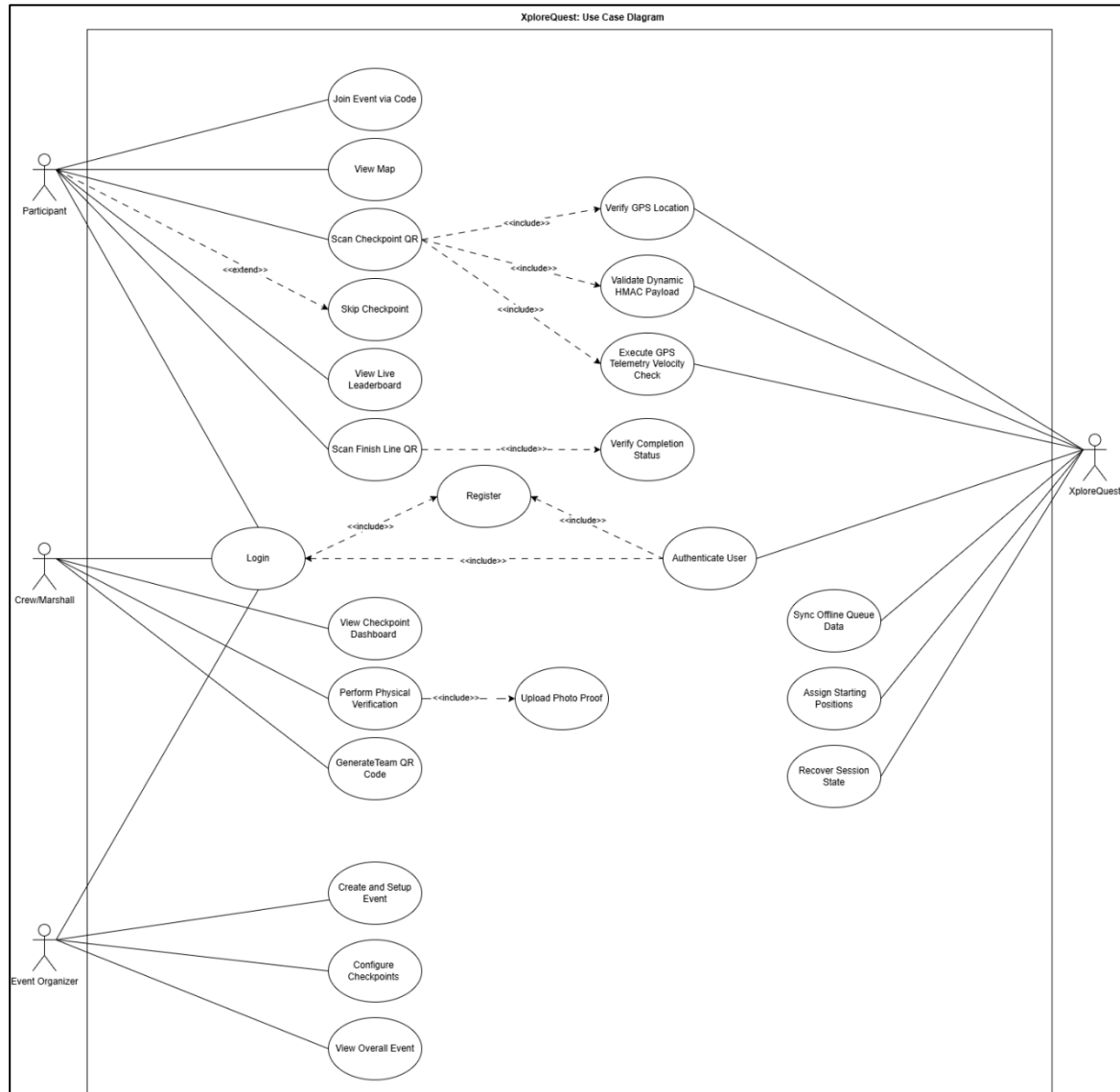


Figure 3 10 Use Case diagram for XploreQuest

- 1 Participant: The team leader who navigates checkpoints, views maps, solves tasks, skips checkpoints, and scans validation QR codes.
- 2 Crew/Marshall: Field staff stationed at checkpoints who monitor arrival queues, perform physical task verifications, capture photo proofs, and generate dynamic team-locked QR codes.
- 3 Event Organizer: Admin who configures events, designs geofence boundaries and monitors live standings.

- 4 XploreQuest (Automated Backend System): Background backend processes that handle GPS checks, validate dynamic HMAC payloads, calculate travel velocity, and sync offline queues.

3.5.3 Flowchart

To visualize these interactions, the workflows for the three user roles and the automated backend are detailed below.

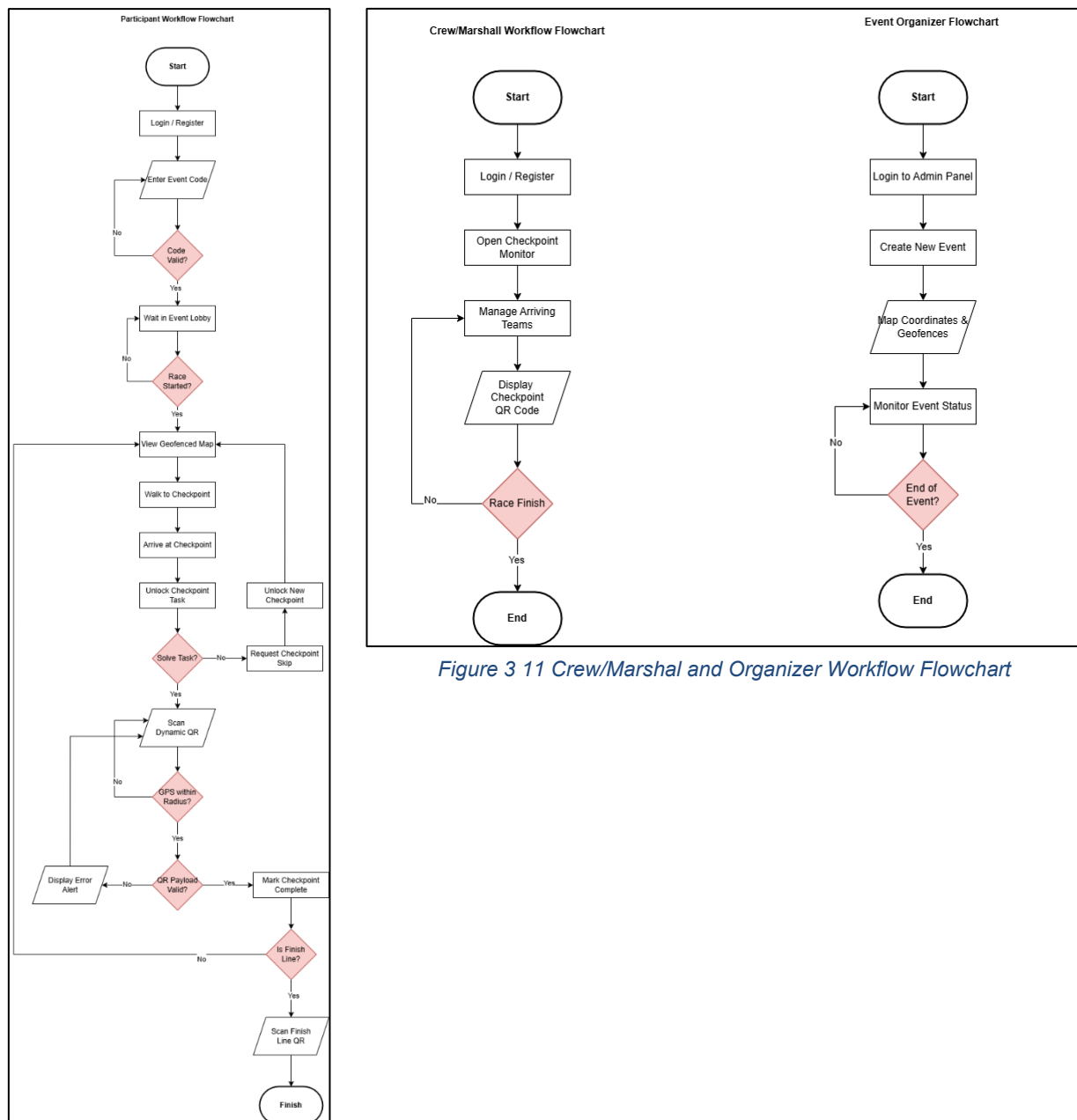


Figure 3 11 Crew/Marshall and Organizer Workflow Flowchart

Figure 3 12 Participant Workflow Flowchart

1. Participant Workflow

The process begins when a participant receives the app and begins to navigate via a geofence map. When a participant arrives at the location of a checkpoint, they scan a dynamically created QR code to have their location validated. Two validation checks are done by the system: 1) GPS coordinates to ensure the participant's GPS coordinates fall within a predetermined radius of the checkpoint and 2) that the participant's unique ID mixed with date/time is stored across multiple secure locations in the HMAC signature. If either test fails, the participant will be denied access to the checkpoint. If both tests are verified, the associated task for that checkpoint will be unlocked for the participant to complete. If the checkpoint is full, the participant can request to skip the checkpoint and will be marked as "Pending". After completing all checkpoints, the participant scans the finish line QR code. The system will validate if all checkpoints were completed (blocking any further check-in attempts if there are pending tasks) and will update the leaderboard.

2. Event Crew Workflow

the Event Crew can view the location of the dynamic QR code that will refresh every 30 seconds. The Event Crew is responsible for monitoring the arrival of teams, validating the completion of physical tasks, uploading photo documentation.

3. Event Organizer Workflow

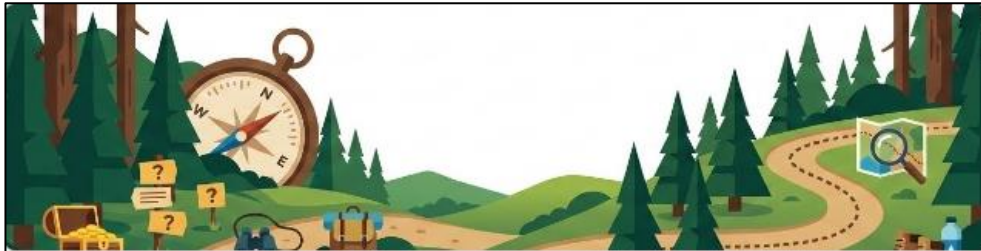
They can create events and setup other specific data for the event such as: definition of coordinates for the event geofence, details on physical tasks & clues, account setup for Event Crew, generate event access code, monitor all safety alerts & current leaderboards in real-time, and export a Race Report at the completion of the event.

3.6 Conclusion

The research and development approach for the XploreQuest system are outlined in Chapter 3. By following the iterative Agile-Scrum model, the developer ensured that the design and implementation sprints are responsive to the stakeholder's feedback. The mixed-methods data collection method, including one-on-one meetings with Puan Nurul Husna (KL Event Crew) to gather qualitative information and the quantitative survey data from 54 respondents, confirmed that there was an urgent need for an ability to perform location verification securely and without Internet connectivity. The functional, non-functional, and hardware/software specifications provide the framework for the design of the system. Use case models and system flowcharts were developed to define the boundaries of the system and how the interactions will take place.

These specifications form the basis of the architectural parameters and will be used in the next phase of the process.

Appendix A – Requirements Specification Document



Survey on User Perception of Secure Mobile Exploration Applications: XploreQuest

Assalamualaikum & Greetings,

I am Muhammad Syah Bin Razak, a student from Universiti Poly-Tech Malaysia (UPTM), currently pursuing my Bachelor's Degree. As part of my Final Year Project (FYP), I am conducting a study on the development of a secure geofencing and exploration mobile application named "XploreQuest".

XploreQuest is designed to facilitate secure, location-based checkpoint validation and progress tracking for outdoor exploration races (Exploraces/Treasure Hunts). This system incorporates dynamic security verifications (time-locked dynamic QR codes and location checks) to prevent common cheating practices like GPS location spoofing and QR screenshot sharing.

This survey takes only 2 to 3 minutes. All responses are anonymous, kept strictly confidential, and used only for academic purposes. Thank you for your time!

[Next](#) [Clear form](#)

Section 1: Demographic Profile

Q1. Age Group *

☐ Under 18

☐ 18 – 24 years

☐ 25 – 30 years

☐ Above 30 years

Q2. Gender *

☐ Male

☐ Female

Q3. Occupation / Status *

☐ Student

☐ Employed

☐ Unemployed

Q4. How often do you participate in outdoor adventure races, scavenger hunts, or exploratory events (e.g., Explorace, running events, treasure hunts)? *

☐ Never

☐ Rarely (1-2 times a year)

☐ Occasionally (3-5 times a year)

☐ Frequently (More than 5 times a year)

Section 2: Experience & Usability Challenges

Q5. Have you ever participated in an outdoor adventure race, Explorace, or treasure hunt event? *

☐ Yes, as a participant.

☐ Yes, as an organizer / crew member.

☐ No, I have never participated.

Q6. During outdoor events (held in parks, forest reserves, or large campuses), how often have you faced cellular network data drops (spotty internet)? *

1 2 3 4 5

Never (Always had excellent network) ☐ ☐ ☐ ☐ ☐ Very Frequently (Faced constant data loss)

Q7. What are the main difficulties or bottlenecks you have encountered during checkpoint-based events? (Select all that apply) *

☐ Long queues and waiting congestion at physical checkpoints

☐ Getting lost or finding checkpoints difficult to navigate to

☐ Staff taking too much time to manually record or sign off team arrivals

☐ Carrying and protecting physical, paper-based stamp cards/clues

☐ Lack of real-time progress tracking or live leaderboards

☐ Network drops making it hard to access event information

☐ I have not faced any issues

☐ Other: _____

Section 3: User Perception of Event Security & Cheating

Q8. Are you aware that participants in digital exploration events can cheat using developer-mode GPS spoofing (mock-location apps) to fake their location? *

☐ Yes, I am aware of this trick.

☐ No, I did not know this was possible.

Q9. Have you ever seen or heard of participants sharing photos/screenshots of checkpoint QR codes via WhatsApp or Telegram so teammates who are far away can check in? *

☐ Yes

☐ No

☐ Not sure

Q10. How unfair do you think it is when participants cheat in a competitive race using these digital methods? *

1 2 3 4 5

Not Unfair at All (It is just a game strategy) ☐ ☐ ☐ ☐ ☐ Extremely Unfair (Ruins the spirit of the competition)

Q11. How important is it that event organizers use an application that actively prevents cheating (like fake GPS and QR code screenshot sharing)? *

1 2 3 4 5

Not Important at All ☐ ☐ ☐ ☐ ☐ Extremely Important

Section 4: Feasibility of Proposed Features (XploreQuest)

Q12. Rate how helpful the following features would be during an outdoor adventure event: ★

XPLOREQUEST

FLEXIBLE ROUTING:
Skip a busy station to complete easier checkpoints first.

OFFLINE-RESILIENT CACHING:
Recording completions offline and auto-syncing when signal returns.

TEAM PHOTO UPLOAD:
Uploading a photo proof at the checkpoint to verify task completion.

DYNAMIC CHECKPOINT CHECK-IN:
Scanning a QR code that changes frequently to ensure physical presence.

VISUAL PROGRESS INDICATOR:
A progress bar and visual checklists (✓) to show race completion.

☐ Not Helpful

☐ Neutral

☐ Very Helpful

Q13. Do you believe that implementing these features in XploreQuest would make outdoor scavenger hunts safer, fairer, and more reliable? ★

☐ Yes, definitely.

☐ Neutral / Not sure.

☐ No.

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Appendix C – Log Book

CT206 / BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN CYBER SECURITY



FACULTY OF COMPUTING & MULTIMEDIA (FCOM)

FINAL YEAR PROJECT 1
FYP4112

LOG BOOK

STUDENT'S NAME : MUHAMMAD SYAH BIN RAZAK

ID NO. : AM2412018244

SUPERVISOR : NUR AFRA BINTI TAIDIN

PROJECT TITLE : A Secure Cross-Platform Mobile Exploration Application

CT206 / BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN CYBER SECURITY

Week		Agenda	Next Agenda	Signature (Supervisor / Coordinator)
11 May – 17 May	1	Brainstormed project Ideas	Formulate project objectives	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia
18 May – 24 May	2	Formulated project objectives	Conduct literature review	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia
25 May – 31 May	3	Conducted literature review	Conduct client consultation	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia
01 June – 07 June	4	Conducted client consultation	Define system specifications	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia
08 June – 14 June	5	Defined system specifications	Design UML diagrams	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia
15 June – 21 June	6	Designed UML diagrams	Prepare presentation slides	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia
22 June – 28 June	7	Prepared presentation slides	Deliver project presentation	NUR AFRA BINTI TAIDIN Lecturer Faculty of Computing and Multimedia Universiti Poly-Tech Malaysia

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